

# Lab Worksheet - Exploratory Data Analysis in airquality

## Dataset

```
# load libraries:
library(tidyverse) # Recall that tidyverse contains dplyr, ggplot2, and many
other useful packages.

# Load `airquality` dataset:
data("airquality")
airquality |>
  head()
```

|   | Ozone | Solar.R | Wind | Temp | Month | Day |
|---|-------|---------|------|------|-------|-----|
| 1 | 41    | 190     | 7.4  | 67   | 5     | 1   |
| 2 | 36    | 118     | 8.0  | 72   | 5     | 2   |
| 3 | 12    | 149     | 12.6 | 74   | 5     | 3   |
| 4 | 18    | 313     | 11.5 | 62   | 5     | 4   |
| 5 | NA    | NA      | 14.3 | 56   | 5     | 5   |
| 6 | 28    | NA      | 14.9 | 66   | 5     | 6   |

Q1: Render this quarto file to a pdf. To do so, you can use the hotkey `Ctrl + Shift + K` or `Cmd + Shift + K`. Then, you will see a command appear in the Positron Terminal. Once the command finishes running, you will see a pdf file appear in the same directory as this file and a VIEWER window will display the pdf on the right pane of Positron. Here you should see the rendered contents of this file.

Q2: Convert the `Month` column to a factor with it's levels labeled as month names (e.g. "May", "June", etc.).

```
# your code here
airquality <- airquality |>
  mutate(Month = factor(Month, levels = 5:9, labels = c("May", "June",
"July", "August", "September")))

airquality |>
  head()
```

|   | Ozone | Solar.R | Wind | Temp | Month | Day |
|---|-------|---------|------|------|-------|-----|
| 1 | 41    | 190     | 7.4  | 67   | May   | 1   |
| 2 | 36    | 118     | 8.0  | 72   | May   | 2   |
| 3 | 12    | 149     | 12.6 | 74   | May   | 3   |
| 4 | 18    | 313     | 11.5 | 62   | May   | 4   |
| 5 | NA    | NA      | 14.3 | 56   | May   | 5   |
| 6 | 28    | NA      | 14.9 | 66   | May   | 6   |

Q3: Create a new column `Temp_C` that converts the temperature from Fahrenheit to Celsius.

```
# your code here
airquality <- airquality |>
  mutate(Temp_C = (Temp - 32) * 5/9)

airquality |>
  head()
```

|   | Ozone | Solar.R | Wind | Temp | Month | Day | Temp_C   |
|---|-------|---------|------|------|-------|-----|----------|
| 1 | 41    | 190     | 7.4  | 67   | May   | 1   | 19.44444 |
| 2 | 36    | 118     | 8.0  | 72   | May   | 2   | 22.22222 |
| 3 | 12    | 149     | 12.6 | 74   | May   | 3   | 23.33333 |
| 4 | 18    | 313     | 11.5 | 62   | May   | 4   | 16.66667 |
| 5 | NA    | NA      | 14.3 | 56   | May   | 5   | 13.33333 |
| 6 | 28    | NA      | 14.9 | 66   | May   | 6   | 18.88889 |

Q4: Rename the `Wind` column to `Wind_mph` and the `Temp` column to `Temp_F`.

```
# your code here
airquality <- airquality |>
  rename(Wind_mph = Wind, Temp_F = Temp)

# or:
# airquality <- airquality |>
#   mutate(Temp_F = Temp, Wind_mph = Wind) |>
#   select(-Temp, -Wind)

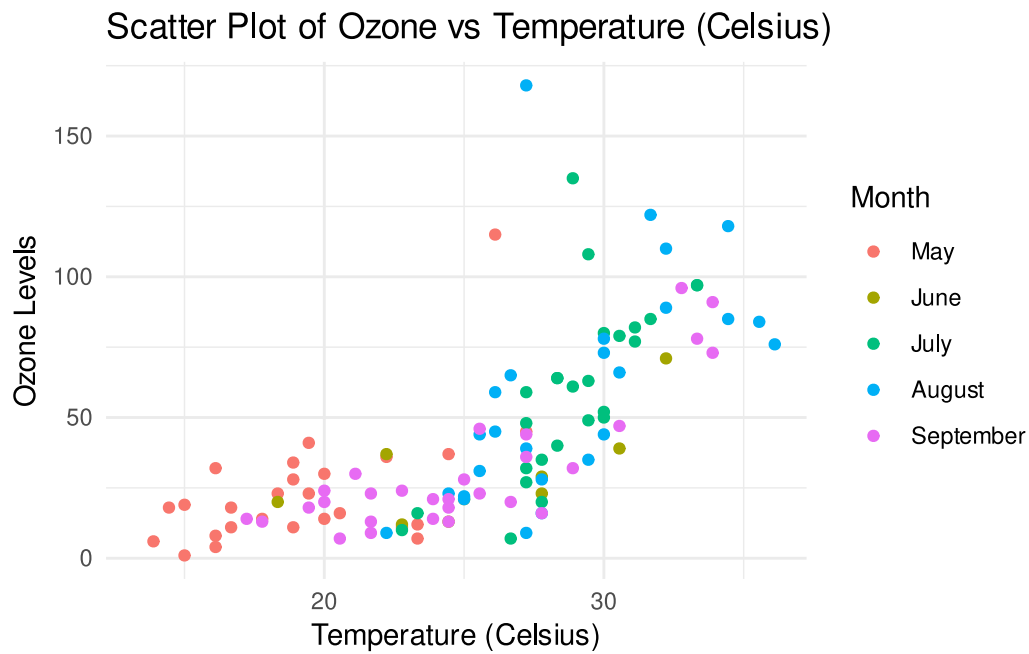
airquality |>
  head()
```

|   | Ozone | Solar.R | Wind_mph | Temp_F | Month | Day | Temp_C   |
|---|-------|---------|----------|--------|-------|-----|----------|
| 1 | 41    | 190     | 7.4      | 67     | May   | 1   | 19.44444 |
| 2 | 36    | 118     | 8.0      | 72     | May   | 2   | 22.22222 |
| 3 | 12    | 149     | 12.6     | 74     | May   | 3   | 23.33333 |
| 4 | 18    | 313     | 11.5     | 62     | May   | 4   | 16.66667 |
| 5 | NA    | NA      | 14.3     | 56     | May   | 5   | 13.33333 |
| 6 | 28    | NA      | 14.9     | 66     | May   | 6   | 18.88889 |

Q5: Create a scatter plot of `Ozone` vs `Temp_C`, colored by `Month`. Add appropriate axis labels and a title.

```
# your code here
airquality |>
  ggplot(aes(x = Temp_C, y = Ozone, color = Month)) +
  geom_point() +
  labs(title = "Scatter Plot of Ozone vs Temperature (Celsius)",
        x = "Temperature (Celsius)",
        y = "Ozone Levels") +
  theme_minimal()
```

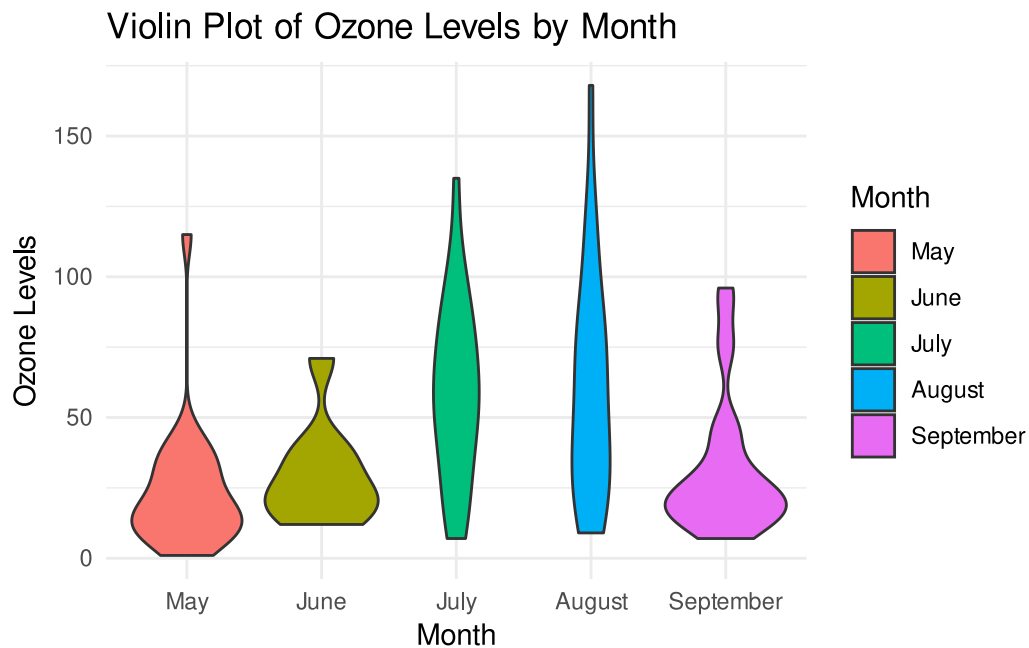
Warning: Removed 37 rows containing missing values or values outside the scale range (``geom_point()``).



Q6: Create a violin plot of `Ozone` levels for each month. Add appropriate axis labels and a title.

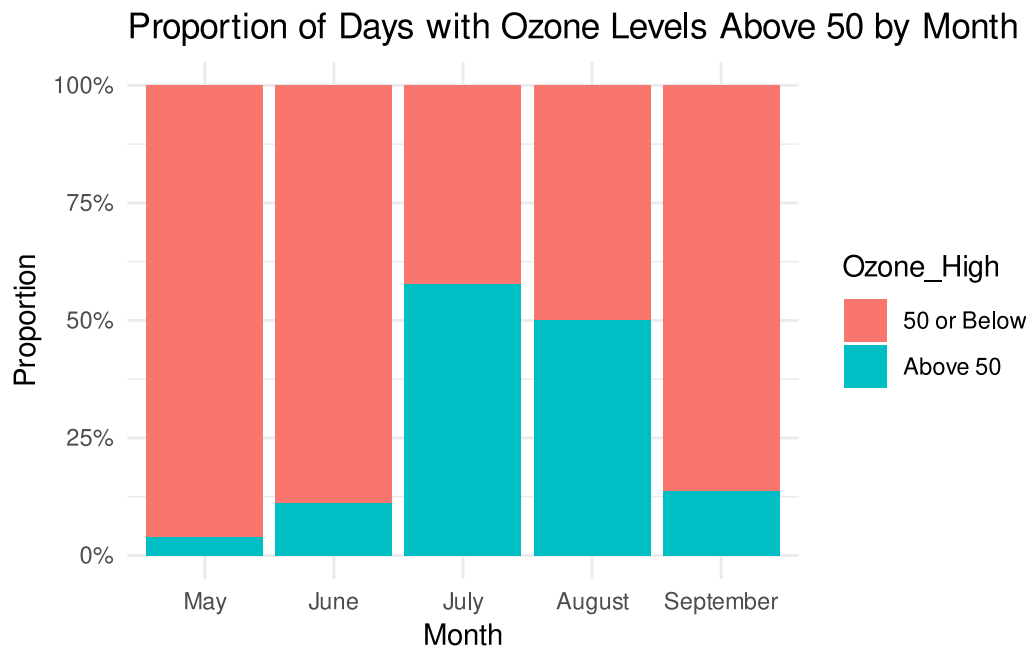
```
# your code here
airquality |>
  ggplot(aes(x = Month, y = Ozone, fill = Month)) +
  geom_violin() +
  labs(title = "Violin Plot of Ozone Levels by Month",
       x = "Month",
       y = "Ozone Levels") +
  theme_minimal()
```

Warning: Removed 37 rows containing non-finite outside the scale range (`'stat_ydensity()'`).

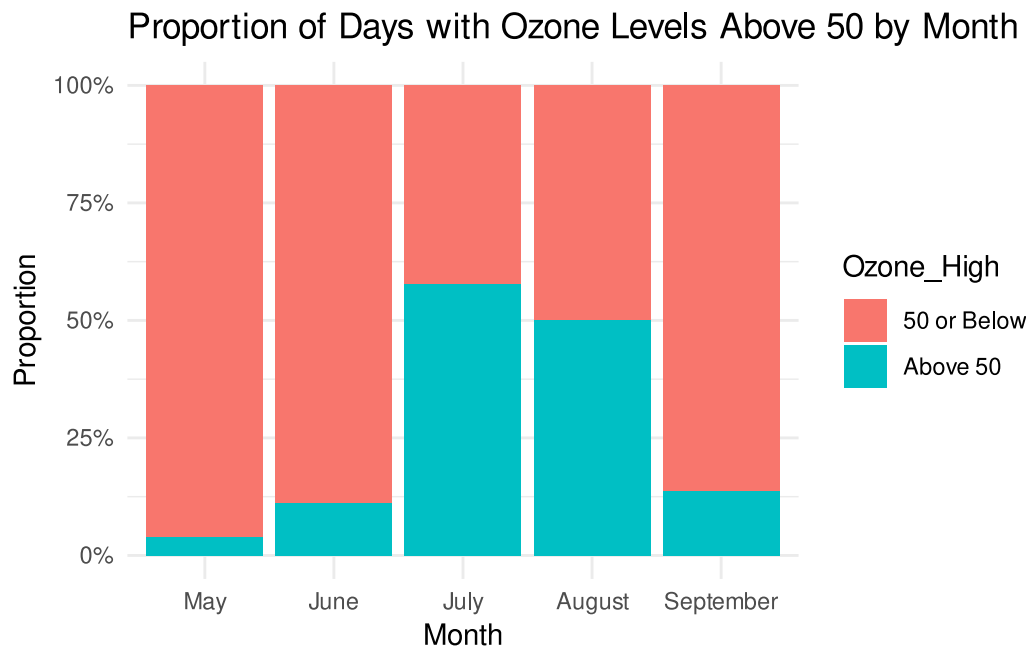


Q7: Create a segmented bar chart of the proportion of days with `Ozone` levels above 50 by `Month`. Remove NA values in the `Ozone` column. Add appropriate axis labels and a title.

```
# your code here
airquality |>
  drop_na(Ozone) |> # or filter(!is.na(Ozone)) |>
  mutate(Ozone_High = ifelse(Ozone > 50, "Above 50", "50 or Below")) |>
  group_by(Month, Ozone_High) |>
  summarise(Count = n(), .groups = 'drop') |>
  group_by(Month) |>
  mutate(Proportion = Count / sum(Count)) |>
  ggplot(aes(x = Month, y = Proportion, fill = Ozone_High)) +
  geom_bar(stat = "identity", position = "fill") +
  labs(title = "Proportion of Days with Ozone Levels Above 50 by Month",
       x = "Month",
       y = "Proportion") +
  scale_y_continuous(labels = scales::percent) +
  theme_minimal()
```

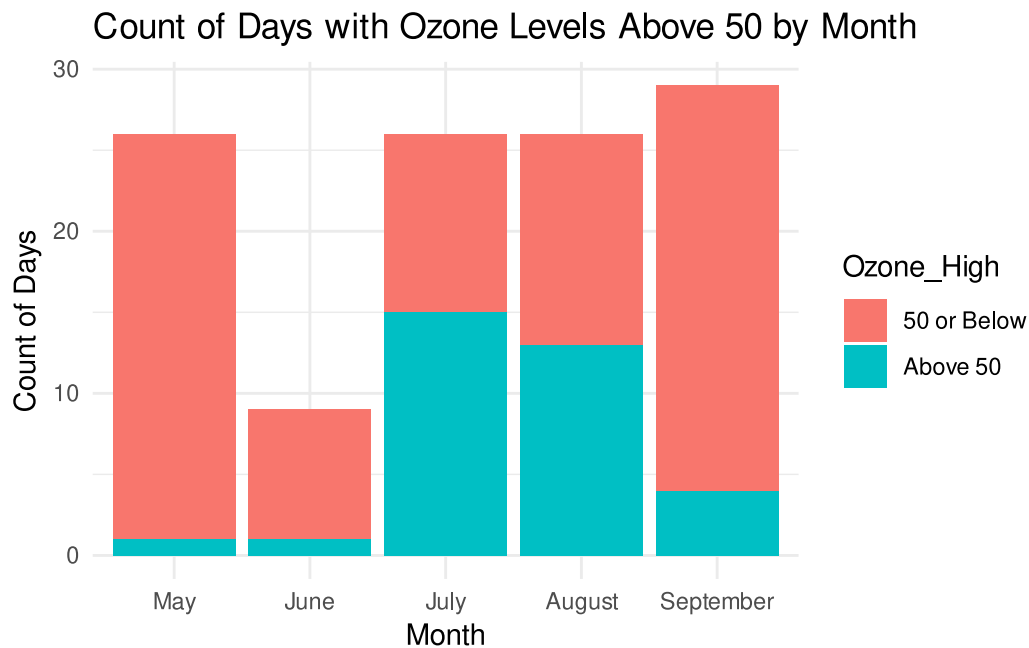


```
# or simply:
airquality |>
  drop_na(Ozone) |> # or filter(!is.na(Ozone)) |>
  mutate(Ozone_High = ifelse(Ozone > 50, "Above 50", "50 or Below")) |>
  ggplot(aes(x = Month, fill = Ozone_High)) + # count is the default
  # statistic in geom_bar, however we fix with position = "fill"
  geom_bar(position = "fill") + # position = "fill" gives proportions
  # instead
  labs(title = "Proportion of Days with Ozone Levels Above 50 by Month",
        x = "Month",
        y = "Proportion") +
  scale_y_continuous(labels = scales::percent) +
  theme_minimal()
```



Q8: Create a stacked bar chart showing the count of days available for each month. Fill bar colors by `Ozone` levels above or below 50 for each month. Remove NA values in the `Ozone` column. Add appropriate axis labels and a title.

```
# your code here
airquality |>
  drop_na(Ozone) |> # or filter(!is.na(Ozone)) |>
  mutate(Ozone_High = ifelse(Ozone > 50, "Above 50", "50 or Below")) |>
  ggplot(aes(x = Month, fill = Ozone_High)) +
  geom_bar() +
  labs(title = "Count of Days with Ozone Levels Above 50 by Month",
       x = "Month",
       y = "Count of Days") +
  theme_minimal()
```



Q9: What is the utility of using a segmented bar chart vs a regular bar chart in this context? Which do you think is more informative and why?

...

Q10: Create a summary table of the number and proportion of NA values in each column.

```
# your code here
airquality |>
  summarize(
    missing_Ozone = sum(is.na(Ozone)),
    prop_missing_Ozone = mean(is.na(Ozone)),
    missing_SolarR = sum(is.na(Solar.R)),
    prop_missing_SolarR = mean(is.na(Solar.R)),
    missing_Wind = sum(is.na(Wind_mph)),
    prop_missing_Wind = mean(is.na(Wind_mph)),
    missing_Temp = sum(is.na(Temp_F)),
    prop_missing_Temp = mean(is.na(Temp_F)),
    missing_Month = sum(is.na(Month)),
    prop_missing_Month = mean(is.na(Month)),
    missing_Day = sum(is.na(Day)),
    prop_missing_Day = mean(is.na(Day))
  )
```



```
) |>
pivot_longer(everything(), names_to = "Metric", values_to = "Value")
```

```
# A tibble: 12 × 2
  Metric      Value
  <chr>      <dbl>
1 missing_Ozone 37
2 prop_missing_Ozone 0.242
3 missing_SolarR 7
4 prop_missing_SolarR 0.0458
5 missing_Wind 0
6 prop_missing_Wind 0
7 missing_Temp 0
8 prop_missing_Temp 0
9 missing_Month 0
10 prop_missing_Month 0
11 missing_Day 0
12 prop_missing_Day 0
```

```
airquality |>
  summarize(across(everything(), ~ sum(is.na(.)))) |>
  pivot_longer(everything(), names_to = "Column", values_to = "Num_NA")
```

```
# A tibble: 7 × 2
  Column Num_NA
  <chr>   <int>
1 Ozone    37
2 Solar.R    7
3 Wind_mph    0
4 Temp_F    0
5 Month    0
6 Day    0
7 Temp_C    0
```

```
airquality |>
  summarize(across(everything(), ~ mean(is.na(.)))) |>
  pivot_longer(everything(), names_to = "Column", values_to = "Prop_NA")
```

```
# A tibble: 7 × 2
  Column Prop_NA
  <chr>   <dbl>
```

|   |          |        |
|---|----------|--------|
| 1 | Ozone    | 0.242  |
| 2 | Solar.R  | 0.0458 |
| 3 | Wind_mph | 0      |
| 4 | Temp_F   | 0      |
| 5 | Month    | 0      |
| 6 | Day      | 0      |
| 7 | Temp_C   | 0      |